General Partitioning Matrix

76 Matrix can be partitioned into sub matrices in various way be ~~intg~~ inserting lines between selected rows and column.

77) ~~metric~~ Addition of Block Matrices A & B the same size and are partitioned in exactly the same way.

78 what is max possible number of Pivots in 6x6 Matrix

→ 6 جتنی row کی اتنے pivot element

79 A is invertible square matrix. then.

→ $(A^{T})^{-1} = (A^{-1})^{T}$

80 LU-decomposition converts the problem solving of single system $Ax=b$ into problem solving of two system $Ly=b$ $Ux=y$ system easy solve because coefficient matrices are ——

⇒ Triangular

81 A Partitioned matrix 'A' said to be block diagonal if the matrices on the main diagonal are square and all other position matrices ——

⇒ Zero

82 If system of equation is solved using Gauss-seidal method, following is best, M desired coffi

all of its entries above the diagonal must zero.

⇒ All

A homogeneous linear system always have the trival solution, there are only two possiblities for solution.

⇒ system has infinitely many solutions in addition to the trival solution..

A = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ following is |A| .

⇒ ad − bc .

(85) An n×n ——— matrix is only square matrix that is both unit lower triangular and upper triangular.

⇒ identity

(86) If the absolutly value of each diagonal entry in a square matrix exceeds the sum of absolute value of other entries in same row.

⇒ It is strictly diagonally dominant matrix

(87) following is diagonal Matrix.

⇒ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ← [illegible] Main diagonal [illegible] non-zero element

(88) Multiplication of partitioned matrix by a scalar is also computed ———

⇒ block by block

(89) Jacobis iteration Converges if a square matrix A is diagonally domenent.

⇒ True.

(90) both the Jacobi and Gauss Sequences Converges for solution of $Ax=b$ for initial $x(0)$,

⇒ Unique solution

(91) Let A be square $n \times n$ matrix. which is equivalent

⇒ A is row equivalent to $n \times n$ identity matrix

(92) How many assumptions are there in Jacobi Method?

⇒ Two ②

(93) Let $Ax=0$ be a homogeneous linear system of 'n' unknown. Then cofficient matrix 'A' if and only if this system has ____ solution.

⇒ ~~infinite many~~ Trival

(94) We can add Matrices of ____

⇒ Same order

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5 If a square matrix A can be reduced to row echelon form with ___ interchange than A has LU-decomposition.

⇒ no-row

56 Non-square Matrices donot have inverse.

⇒ True

(57) which one is null Matrix

⇒ $\begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$

(58) true partitioned matrices $Y=[Q\ R]$ $Z=[U\ V]$ sub matrices Q and R have same size as U & V.

⇒ $Y+Z=[Q+U \quad R+V]$

59 For non singular matrix A, A^{1} equal to

⇒ $\frac{1}{\det(A)} * Adj(A)$ or ($Adj(A)/\det(A)$)

(160 A Partitioned square matrix 'A' is said to be ___ matrix. if the matrices on the main diagonal are squared all materices below ~~are~~ the ~~square~~ main diagonal are zero.

⇒ block upper triangular.

(161 A particular square matrix 'A' said to be ___ matrix on the main diagonal are square and all matrices above the main diagonal

⇒ block lower triangular

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(102) $T : R^2 \to R^2$ be transformation $T(x_1, x_2) = (x, 0)$ The null space (or kernel) $N(T)$ of T is

⇒ $(0, x_2)$

(103) matrix of A order 2×3 and B of order 3×5? AB

⇒ 2×5

(104) Jacobi's method, rate of convergence is quite ⟵ Compared other methods.

⇒ slow

(105) A system of linear equation is said to be homogeneous if it can be written in form —

⇒ $AX = 0$

(106) true speed of convergence of both the Jacobi and Gauss Seidal sequance?

⇒ depends on the diagonal entries dominate the corresponding row sums.

(107) A & B are square matrix. A & B are invertible with $B = A^{-1}$, $A = B^{-1}$ if and only if $AB = BA$ equal to — matrix.

⇒ Identity

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Consider ~~and~~ an LU-factorization, the matrix L is ___ and is called a unit lower triangular Matrix.

⇒ Invertible

(109- If A is strictly diagonally matrix, following is true.

⇒ It is invertible matrix

(110 Jacobic method, diagonals entries of matrix are ___

⇒ non-zero

(111 homogeneous linear system always has the trivial solution. there are only two possibilities.

⇒ system has infinitely many solution in addition to trivial solution.

(112 A decomposition of matrix as a product of two or more matrics is called Matric ___.

⇒ factorization

(113) what is possible number of povids in 3by3 Matrix.

⇒ 3 جتنی Row ہوں گی اتنے ہی آئیں گے

(114 max possible number of povids in 4×6 matrix

⇒ 4

(115 Let T be a linear transformation and A be a standard matrix of T. T is invertible if and only if A is ___ Matrix.

→ invertible

57 Power of

Transpose Matrix an $m \times n$ matrix A transpose of A is $n \times m$ matrix. A^t denoted.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 5 & 6 \end{bmatrix} \Rightarrow A^t = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 6 \end{bmatrix}$$

58 (1) $(A^t)^t = A$ (3) $(rA)^t = rA^t$

(2) $A+B)^t = A^t + B^t$ (4) $(AB)^t = B^t A^t$

59 Transpose of Product of matrices equal to product of their transposes in the reverse order.

60 Inverse of of square Matrix if A is $n \times n$ matrix A matric C of order $n \times n$ is Called multiplicative inverse of A. $\boxed{AC = CA = I}$ identity Matrix

61 If inverse of square matrix exists, it is Called invertible matrix

62 $AA^{-1} = I$ $A^{-1}A = I$

63 Matrix that is not invertible is sometimes Called singular matrix. → $|A| = 0$

64 Invertible matrix is Called non-singular Matrix ← $|A| \neq 0$

65 $A^{-1} = \frac{1}{|A|} Adj(A)$ $(A^{-1})^{-1} = A$

66 If A is invertible matrix then A^{-1} is also invertible

67 A and B are $n \times n$ invertible matrices so AB so the inverse of AB is product of inverse of A & B reverse order. $(AB)^{-1} = B^{-1}A^{-1}$

Assume // Row اگر [illegible] . [illegible]

a_{11} → Column
↓
Row

46 **Diagonal Matrix** is Square matrix who non-diagonal entries are Zero [illegible] $\begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 11 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 16 & 0 \end{bmatrix}$

47 **Null Matrix or Zero Matrix** who all entries are zero or null.

48 **Equal Matrix** have the same size $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ $\begin{vmatrix} 2 & 1 & 0 \end{vmatrix}$

49 & Corresponding entries

49 during adding or subtracting order will be same.

50 Ex $\begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \end{bmatrix}$ & $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$ not defined because they have diff size.

51 **Properties (Addition)**

1) $A+B = B+A$ ④ $r(A+B) = rA + rB$

2) $(A+B)+C = A+(B+C)$ ⑤ $(r+s)A = rA + sA$

3) $A+0 = A$ ⑥ $r(sA) = (rs)A$

52 Matrix Multiplication (its row and Column are equal then apply multiply)

53 $\underline{A = \overset{2\times3}{m \times n}}$ $\underline{B = \overset{3\times4}{n \times p}}$ $\underline{AB = ?}$ $\Rightarrow \overset{2\times4}{m \times p}$

54 matrix B has size $\underline{3\times4}$ and A has size $\underline{2\times3}$. inside number are not same, so product BA undefined

55 If A is $\underline{3\times5}$ and B is $\underline{5\times2}$ matrix what is sizes of AB and BA. If they defined?

$\Rightarrow \underline{AB = 3\times2}$ $B \cdot A \Rightarrow$ not defined $\underline{B = 5\times\overset{\text{Column}}{\overset{\uparrow}{2}}}$. $\underline{A = \overset{\text{Row}}{\overset{\uparrow}{3}}\times5}$ not equal

56 **Properties (Multiplication)**

(i) $A(BC) = (AB)C$ Associative law of multiplication

(ii) $A(B+C) = AB+AC$ left distribution law

(iii) $(B+C)A = BA+CA$ Right di " "

(iv) $r(AB) = (rA)B + A(rB)$ for any scalar r

(v) $I_m A = A = A I_m$ identity Matric multiplication

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Quiz#01

33 Linear Transformation $A = [T(e_1)\ T(e_2) \cdots T(e_n)]$

$T(x) = x_1 T(e_1) + x_2 T(e_2)$

34 $A = [T(e_1) \ldots T(e_n)]$

35 every Linear Transformation from R^n to R^m is matrix Transformation

36 Linear Transformation focus on Property of mapping.

37 Dilation Transformation $T(x) =$ ____

$\Rightarrow 3x$ — Dilation Trm اگر x کے ساتھ کوئی رقم آتی ہے تو یہ

38 if $R^3 \to R^3$ then standard Matrix $A = [T(e_1)\ T(e_2)\ T(e_3)]$

39 mapping $T: R^n \to R^m$ is said to be one to one (or 1:1) each b in R^m is the image of at most one x in R^n.

40 mapping $T: R^n \to R^m$ is said to be onto R^m if each b in R^m is the image of at least one x in R^n.

41 mapping T is not one to one when some b in R^m is the image of more than one vector in R^n.

42 $T(x) = Ax =$ ____

$\Rightarrow T(x) = Ax = b$

43 Kernel of Linear Transformation = Ker T

44 $\text{Ker}T = \{v = V / T(v) = 0 \text{ in } W\}$

Set of those elements in 'V' which map onto the zero vector in W.

45 if T is one to one iff Ker [illegible] in V